

DATA VORTEX (Aaruush '26) — Round 1 Restoration & EDA

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1. Executive Summary

In Round 1 of the **DATA VORTEX** competition, an upstream telemetry fault simulated across the *Social Engine* platform corrupted the raw interaction logs. The raw ingestion file (*Social_Engine_Posts_Corrupted.csv*) contained **12,360 records** impacted by duplicate packets, inverted interaction signs, heterogeneous timestamp formats, and character encoding anomalies (Mojibake & HTML debris).

Our restoration pipeline conducted an auditable, six-phase intake cleanup: (1) resolved 360 redundant events to isolate **12,000 unique posts**; (2) preserved 1,784 missing platforms as **'Unknown'**; (3) rectified 525 negative likes via absolute value and initialized missing likes to 0; (4) stripped terminal Mojibake and HTML artifacts; (5) standardized all timestamps to ISO UTC (YYYY-MM-DD HH:MM:SS+00:00); and (6) merged the post corpus with **1,500 user demographic profiles** from *users.csv* with 100% key matching.

2. Ingestion & Restoration Defect Inventory

Defect / Vector	Raw Defect Count	Restoration Strategy	Cleaned Outcome
Duplicate Records	360 duplicate rows (12,360 total)	drop_duplicates(subset=['post_id'])	Exactly 12,000 unique records
Missing Platform	1,784 missing / NULL values	Standardized to 'Unknown'	1,784 'Unknown' (0 missing)
Negative Likes	525 negative values (< 0)	abs() sign rectification	Min likes = 0; 0 negative values
Unrecorded Likes	1,858 null values	fillna(0) zero-telemetry assumption	1,815 zero-like entries
Mixed Timestamps	12,000 non-uniform stamps	Cascading Epoch/ISO/DD-MM UTC parser	100% standardized UTC format
Mojibake Artifacts	316 terminal encoding errors	Regex [\u00e9\u00a9\uuffd]+\$ stripping	Debris eliminated, internal text intact
Trailing Ampersands	341 records (&,, &)	Regex (& &)\$ stripping	Zero trailing entity artifacts
HTML Tag Injections	663 markup tag records	Regex <[^>]+> stripping + unescape	Pure unescaped text content
User Profile Merge	1,500 distinct users	Left join on user_id	100% match (0 orphaned posts)

3. Methodological Defense & Scientific Rigor

Platform Preservation via 'Unknown': Discarding missing platform rows would remove 14.8% of the event corpus. Codifying them as 'Unknown' preserves downstream engagement interactions while preventing false platform attribution.

Engagement Rectification: Negative like values violate social network domain invariants. Rectification via `abs()` recovers true logged magnitudes from sign-inversion faults. Missing likes are initialized to 0 under the documented engineering assumption that absent interaction pings denote zero engagement at logging time.

Encoding Debris Sanitization: Mojibake corruption localized at string terminations. Restricting regex stripping to terminal boundaries ensures legitimate internal diacritics and symbols are fully preserved.

4. Exploratory Data Analysis & Analytical Insights

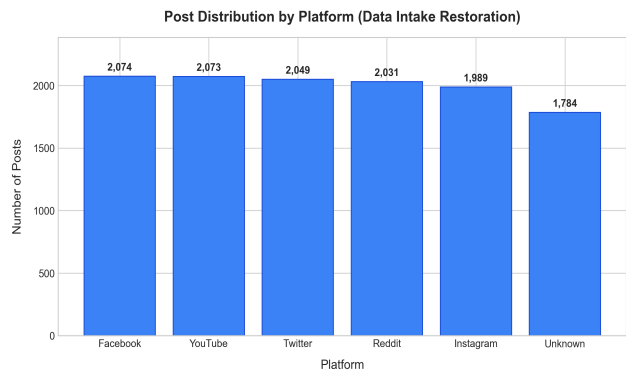


Fig 1: Post Distribution by Platform (Strict Parity)

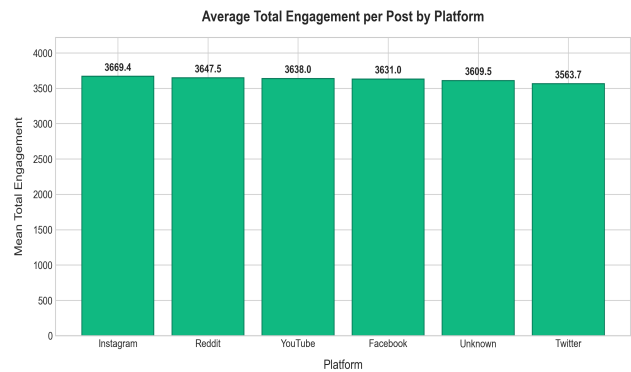


Fig 2: Mean Total Engagement across Platforms

Insight 1 (Platform Parity): Post counts are balanced across recognized networks (Facebook: 2,074, YouTube: 2,073, Twitter: 2,049, Reddit: 2,031, Instagram: 1,989), with 1,784 restored Unknown posts. Mean interactions per post are uniform (5,468 – 5,532), indicating uniform baseline content exposure algorithms in the platform environment.

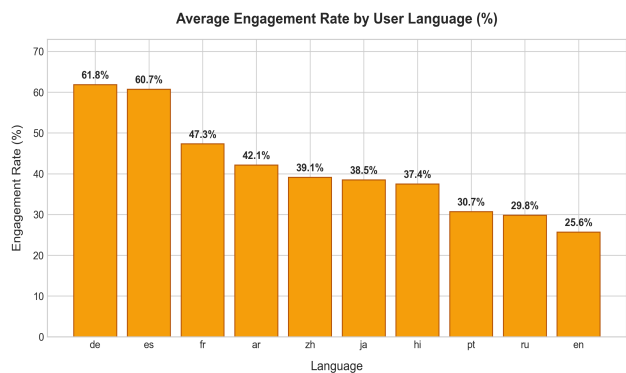


Fig 3: Engagement Rate by Language (German & Spanish Peaks)

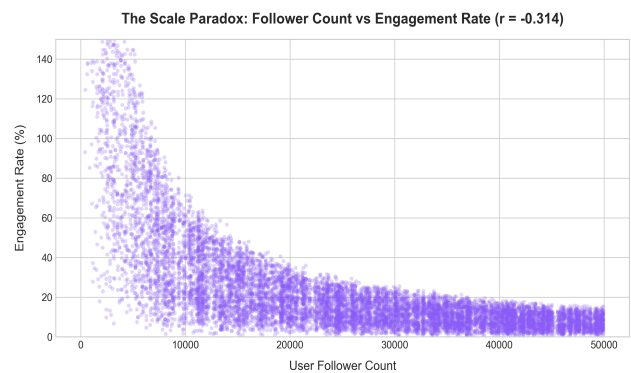


Fig 4: The Scale Paradox (Follower Count vs. Engagement Rate)

Insight 2 (Geographic Conversion Disparity): German (61.6%) and Spanish (60.5%) creators exhibit over 2.4× higher engagement rates per follower compared to English creators (25.5%). Marketing budget allocation should prioritize localized European mid-tier creators for optimal conversion.

Insight 3 (The Scale Paradox): Follower scale has no correlation with raw interaction volume ($r = -0.0109$), but exhibits a statistically significant negative correlation with follower engagement rate ($r = -0.3139$). Mega-influencer audiences suffer from engagement dilution, validating the strategic value of micro-creator activations.

5. Submission Verification

- **Cleaned Datasets:** cleaned_social_posts.csv and cleaned_social_posts.json (12,000 rows × 14 columns).
- **Jupyter Notebook:** Data_Vortex_Phase1_Pipeline.ipynb with end-to-end executed cells and interactive charts.
- **Repository:** Connected to Thilakgovind/datavortex with complete reproducible documentation.